

Chapter 9E

Demyelinating, Neuromuscular, Spinal Cord, Neurocutaneous, and Vertigo Disorders

Medvale Internal Medicine - Neurology for Internal Medicine

This chapter uses the remaining uploaded neurology segments as coverage guides: multiple sclerosis, Guillain-Barre syndrome, myasthenia gravis, neurocutaneous syndromes, spinal cord diseases, and vertigo. The goal is to connect localization, tempo, danger signs, diagnostic studies, and first-line management into a single internal-medicine framework.

1. Big picture: localize the system before naming the disease

The remaining neurology topics are easiest when organized by anatomic compartment. Multiple sclerosis is a central nervous system demyelinating disease. Guillain-Barre syndrome is an acute peripheral nerve polyradiculoneuropathy. Myasthenia gravis is a neuromuscular junction transmission disorder. Syringomyelia, Brown-Sequard syndrome, transverse myelitis, and poliomyelitis are spinal cord or anterior horn cell syndromes. Vertigo may come from the vestibular apparatus or the brainstem/cerebellum. Neurocutaneous syndromes are genetic or developmental disorders that link skin findings to CNS tumors, seizures, and structural complications.

The clinical reasoning sequence is: first decide whether the deficit is central or peripheral; then decide whether it is monophasic, relapsing, progressive, fluctuating, or episodic; then look for red flags that require respiratory monitoring, urgent MRI, lumbar puncture, neurosurgical evaluation, immunotherapy, or stroke evaluation. A patient with ascending weakness may need pulmonary function monitoring before the diagnosis is fully confirmed. A patient with acute myelopathy needs spine MRI quickly. A patient with vertigo plus brainstem signs needs stroke evaluation, not meclizine alone.

Compartment-first framework

Compartment	Typical disorders	Core pattern	Danger signal
CNS white matter	Multiple sclerosis, neuromyelitis optica spectrum disorder, MOG-associated disease, PML.	Relapsing focal neurologic deficits separated in time and space; optic neuritis, internuclear ophthalmoplegia, sensory/motor episodes.	Spinal cord syndrome, optic neuritis with poor recovery, severe brainstem symptoms, immunosuppression.
Peripheral nerves/roots	Guillain-Barre syndrome, CIDP, diabetic neuropathy.	Ascending weakness, areflexia, sensory symptoms, autonomic instability.	Respiratory failure, bulbar weakness, arrhythmia, rapid progression.
Neuromuscular junction	Myasthenia gravis, Lambert-Eaton syndrome, botulism.	Fluctuating fatigable weakness, ocular/bulbar symptoms, preserved sensation.	Myasthenic crisis, dysphagia, aspiration, low forced vital capacity.
Spinal cord	Syringomyelia, Brown-Sequard syndrome, transverse myelitis, compressive myelopathy.	Sensory level, bilateral weakness, sphincter dysfunction, tract-specific dissociation.	Acute cord compression or urinary retention.
Vestibular/brainstem	BPPV, vestibular neuritis, Meniere disease, cerebellar/brainstem stroke.	Vertigo, nystagmus, imbalance, hearing symptoms or focal neurologic signs.	Central nystagmus, severe gait ataxia, diplopia, dysarthria, dysphagia.
Skin-neuro syndromes	NF1, NF2, tuberous sclerosis, Sturge-Weber.	Cafe-au-lait spots, tumors, seizures, developmental findings, vascular malformations.	New neurologic deficit, vision/hearing loss, seizures, tumor compression.

2. Multiple sclerosis and CNS demyelination

Multiple sclerosis is an immune-mediated demyelinating disease of the central nervous system. It affects brain, optic nerves, brainstem, cerebellum, and spinal cord white matter. The classic diagnostic idea is dissemination in time and space: neurologic

lesions affecting different CNS locations at different times, with no better explanation. Typical presentations include optic neuritis, internuclear ophthalmoplegia, focal sensory symptoms, focal weakness, ataxia, bladder dysfunction, Lhermitte phenomenon, and heat-related worsening of symptoms.

The uploaded source emphasizes young adults with relapsing-remitting symptoms, optic neuritis, internuclear ophthalmoplegia, cerebellar signs, bladder dysfunction, autonomic issues, and MRI/CSF findings. The practical teaching point is that MS is not diagnosed from one vague neurologic episode; the pattern must fit objective CNS lesions separated by time and space, supported by MRI and sometimes CSF oligoclonal bands.

Multiple sclerosis clinical patterns

Finding	Localization	Clinical clue
Optic neuritis	Optic nerve	Pain with eye movement, monocular vision loss, central scotoma, decreased color vision.
Internuclear ophthalmoplegia	Medial longitudinal fasciculus	Impaired adduction of one eye with abducting nystagmus of the other; strongly suggests demyelination in young adult.
Lhermitte phenomenon	Cervical spinal cord dorsal columns	Electric shock-like sensation down spine with neck flexion.
Sensory/motor relapse	CNS white matter tract	Symptoms develop over hours to days and last more than 24 hours.
Cerebellar involvement	Cerebellum / cerebellar peduncles	Ataxia, intention tremor, dysarthria, nystagmus.
Bladder dysfunction	Spinal cord pathways	Urgency, retention, incontinence, recurrent UTIs.

MS diagnostic and treatment framework

- MRI brain and spinal cord with contrast is the key diagnostic test when MS is suspected.
- CSF oligoclonal bands support diagnosis, especially when MRI and clinical criteria need additional evidence.
- Evoked potentials may show delayed conduction through demyelinated pathways, but are not usually first-line in modern workups.
- High-dose corticosteroids can shorten acute relapse duration but do not replace disease-modifying therapy.
- Disease-modifying therapy is used to reduce relapse activity and long-term disability risk; selection is individualized by relapse severity, MRI activity, comorbidities, pregnancy plans, infection risk, and monitoring burden.
- Avoid diagnosing relapse when symptoms are from fever, infection, heat, or metabolic stress causing pseudoexacerbation.

MS trap: a urinary tract infection or fever can transiently worsen old MS symptoms without representing a new inflammatory relapse. Check for infection, heat exposure, and metabolic triggers before escalating steroids.

MS versus important mimics

Mimic	Why it matters	Clues against typical MS
Neuromyelitis optica spectrum disorder	Can cause severe optic neuritis and longitudinally extensive transverse myelitis; treatment differs.	AQP4-IgG, severe bilateral optic neuritis, area postrema symptoms, long cord lesion.
MOG-associated disease	Can resemble MS, optic neuritis, ADEM, transverse myelitis.	MOG-IgG, different relapse pattern and MRI features.
B12 deficiency	Can cause myelopathy and neuropathy.	Macrocytosis, low B12, dorsal column signs, peripheral neuropathy.
Vasculitis / autoimmune disease	Can cause multifocal CNS lesions.	Systemic rash, arthritis, renal disease, inflammatory markers.
PML	JC virus demyelination in immunosuppressed patients.	Subacute progressive deficits, immunosuppression, characteristic MRI, CSF JC virus.

3. Guillain-Barre syndrome

Guillain-Barre syndrome is an acute inflammatory demyelinating polyradiculoneuropathy in its classic form. It usually follows infection and presents with rapidly progressive, often ascending weakness, areflexia or hyporeflexia, mild sensory symptoms, pain, and autonomic instability. Campylobacter jejuni, CMV, EBV, hepatitis, HIV, surgery, and other immune triggers may precede the syndrome. The source emphasizes that weakness may progress over days, respiratory failure can occur, and sphincter function and cognition are usually preserved.

The most important management decision is respiratory and autonomic monitoring. GBS can look like a neurologic diagnosis but behave like an ICU disease. Check forced vital capacity, negative inspiratory force, bulbar symptoms, cough strength, secretion handling, oxygenation, and autonomic instability. Do not rely on oxygen saturation alone, because hypercapnic respiratory failure can develop late after mechanics fail.

GBS pattern recognition

Feature	Typical GBS	Red flag / alternative
Weakness	Ascending symmetric weakness, proximal and distal, evolving over days to weeks.	Marked asymmetry suggests alternative diagnosis.
Reflexes	Decreased or absent reflexes.	Brisk reflexes suggest CNS lesion or myelopathy.
Sensation	Pain and paresthesias common; objective sensory loss may be mild.	Clear sensory level suggests spinal cord lesion.
Autonomic	Tachycardia, arrhythmia, BP swings, urinary retention, ileus.	Can be life-threatening.
Respiratory	Diaphragm and bulbar weakness may occur.	Declining FVC/NIF, dysphagia, weak cough, inability to count on one breath.
CSF	Elevated protein with normal or low cell count after several days.	High WBC count suggests infection, malignancy, HIV, Lyme, or other process.

GBS diagnosis and treatment

- Diagnosis is clinical, supported by CSF albuminocytologic dissociation and nerve conduction studies showing demyelinating or axonal neuropathy.
- CSF protein may be normal early; a normal early LP does not exclude GBS.
- Hospitalize suspected GBS because respiratory failure and autonomic instability can evolve quickly.
- Treat significant GBS with IVIG or plasma exchange; both can hasten recovery, but they are not usually combined sequentially because benefit is not established.
- Corticosteroids are not helpful for classic GBS.
- Provide DVT prophylaxis, pain control, bowel/bladder care, nutrition, pulmonary monitoring, and rehabilitation.

GBS trap: if a patient has a sensory level, early bowel/bladder dysfunction, brisk reflexes, or severe back pain with focal cord signs, think spinal cord disease rather than peripheral polyradiculoneuropathy.

4. Myasthenia gravis and neuromuscular junction disease

Myasthenia gravis is an autoimmune neuromuscular junction disorder, most often caused by antibodies against the postsynaptic acetylcholine receptor. It produces fluctuating fatigable weakness that worsens with repeated use and improves with rest. Sensation and reflexes are preserved. Ocular symptoms such as ptosis and diplopia are common early manifestations. Bulbar symptoms include dysarthria, dysphagia, chewing fatigue, nasal speech, and aspiration risk. Respiratory muscle weakness produces myasthenic crisis.

The uploaded source highlights myasthenic crisis as a medical emergency, ocular and bulbar involvement, acetylcholine receptor antibody testing, decremental response on repetitive nerve stimulation, thymoma association, pyridostigmine, thymectomy, immunosuppression, plasmapheresis, IVIG, and medications that worsen symptoms. The internal medicine priority is recognizing crisis before hypoxia occurs; forced vital capacity declines before oxygen saturation falls.

Neuromuscular junction comparison

Disorder	Weakness pattern	Autonomic / pupils	Key tests	Treatment anchor
Myasthenia gravis	Fatigable ocular, bulbar, limb, or respiratory weakness; worse with use.	Pupils spared; sensation/reflexes preserved.	AChR/MuSK antibodies, EMG/repetitive stimulation, chest imaging for thymoma.	Pyridostigmine, immunotherapy, thymectomy in selected patients, IVIG/PLEX for crisis.

Disorder	Weakness pattern	Autonomic / pupils	Key tests	Treatment anchor
Lambert-Eaton syndrome	Proximal leg weakness that may improve briefly with exertion.	Dry mouth, autonomic symptoms; reflexes reduced.	P/Q-type voltage-gated calcium channel antibodies; search for small cell lung cancer.	Treat cancer, amifampridine, immunotherapy.
Botulism	Descending weakness, diplopia, dysphagia, dry mouth; pupils may be involved.	Prominent autonomic symptoms and dilated/fixed pupils.	Clinical plus toxin testing; public health notification.	Antitoxin, respiratory support.

Myasthenic crisis

- Suspect crisis with dyspnea, orthopnea, weak cough, dysphagia, pooling secretions, nasal speech, rapidly worsening bulbar weakness, or low FVC/NIF.
- Do not wait for oxygen desaturation; ventilatory mechanics fail before pulse oximetry falls.
- Hold or avoid medications that worsen MG when possible: aminoglycosides, fluoroquinolones, macrolides, beta-blockers, magnesium, some antiarrhythmics, and neuromuscular blockers.
- Treat crisis with ICU monitoring, respiratory support, IVIG or plasma exchange, and management of triggers such as infection or medication exposure.
- Pyridostigmine helps symptoms but may be held during intubated crisis if secretions are problematic; specialist guidance is needed.

5. Spinal cord syndromes

Spinal cord disease often announces itself through a sensory level, bilateral long-tract signs, sphincter dysfunction, and combinations of motor, pain/temperature, vibration/proprioception, and reflex changes. MRI is the key test when acute myelopathy is suspected. The uploaded source includes syringomyelia, Brown-Sequard syndrome, transverse myelitis, and poliomyelitis; the shared concept is tract localization.

Spinal cord tract localization

Tract / structure	Function	Lesion clue
Corticospinal tract	Motor control below lesion.	Weakness, spasticity, hyperreflexia, Babinski below chronic lesion.
Dorsal columns	Vibration, proprioception, fine touch.	Sensory ataxia, positive Romberg, loss of position/vibration.
Spinothalamic tract	Pain and temperature.	Contralateral pain/temp loss below lesion after crossing.
Anterior horn cells	Lower motor neuron output.	Flaccid weakness, atrophy, fasciculations, reduced reflexes.
Central cord crossing fibers	Pain/temp fibers crossing anterior commissure.	Cape-like bilateral pain/temp loss in syringomyelia.
Autonomic/sacral pathways	Bladder, bowel, sexual function.	Urinary retention/incontinence, sphincter dysfunction.

Specific spinal cord syndromes

Syndrome	Classic pattern	Diagnosis / treatment anchor
Syringomyelia	Central cavitation, often cervical; cape-like bilateral loss of pain/temp over shoulders/arms with preserved touch early; hand weakness may occur.	MRI; evaluate for Chiari malformation, tumor, tethered cord, trauma; neurosurgery if symptomatic/progressive.
Brown-Sequard syndrome	Hemisection: ipsilateral weakness and vibration/proprioception loss, contralateral pain/temp loss below lesion.	Often trauma, tumor, abscess; MRI and cause-directed treatment.
Transverse myelitis	Rapid bilateral weakness/sensory deficits below a level plus sphincter dysfunction; often thoracic.	MRI with contrast, CSF, inflammatory/infectious workup; high-dose steroids often used.
Poliomyelitis	Anterior horn cell disease causing asymmetric flaccid weakness with normal sensation; bulbar involvement can impair respiration.	Prevention by vaccination; supportive treatment.
Compressive myelopathy	Back pain, weakness, sensory level, hyperreflexia, urinary retention.	Urgent MRI and neurosurgical/radiation/infectious management depending on cause.

Acute myelopathy trap: urinary retention plus bilateral leg weakness or a sensory level is a spinal cord emergency until proven otherwise. Do not attribute it to peripheral neuropathy without imaging.

6. Vertigo: peripheral versus central

Vertigo is an illusion of movement, usually spinning. The uploaded source emphasizes differentiating peripheral vertigo from central vertigo. Peripheral vertigo is often intense, episodic, associated with nausea/vomiting, worsened by head position, and may include hearing symptoms. Central vertigo is more likely when onset is gradual or persistent, nystagmus is vertical or direction-changing, neurologic deficits are present, or the patient cannot stand or walk.

The HINTS exam can help distinguish vestibular neuritis from posterior circulation stroke in a patient with continuous acute vestibular syndrome, but it requires expertise and is not for brief positional vertigo. In high-risk or unclear cases, especially with focal neurologic signs or inability to walk, MRI/vascular imaging and stroke evaluation are appropriate.

Peripheral versus central vertigo

Feature	Peripheral vertigo	Central vertigo
Common causes	BPPV, vestibular neuritis, Meniere disease, labyrinthitis, ototoxicity.	Cerebellar/brainstem stroke, MS, tumor, vestibular migraine.
Nystagmus	Usually unidirectional horizontal-torsional; suppressed by fixation.	Vertical, direction-changing, pure torsional, or not suppressed.
Neurologic deficits	Absent except hearing symptoms in selected disorders.	Diplopia, dysarthria, dysphagia, ataxia, weakness, numbness.
Gait	Unsteady but often can walk with support.	Severe truncal ataxia or inability to walk is concerning.
Hearing/tinnitus	May occur in Meniere/labyrinthitis/acoustic neuroma.	Usually absent in brainstem/cerebellar stroke, though AICA stroke can affect hearing.
Imaging need	Often none for classic BPPV.	MRI/vascular imaging when central signs or high-risk presentation.

Common vertigo syndromes

Syndrome	Clinical script	Treatment frame
BPPV	Brief episodic positional vertigo triggered by head movement; no hearing loss; fatigable positional nystagmus.	Canalith repositioning maneuvers; avoid chronic vestibular suppressants.
Vestibular neuritis	Acute continuous vertigo for days, nausea, gait instability, no hearing loss.	Short course vestibular suppressants/antiemetics, vestibular rehab; consider steroids in selected cases.
Labyrinthitis	Vertigo plus hearing loss after viral/bacterial inner ear inflammation.	Supportive and cause-directed therapy.
Meniere disease	Episodic vertigo, fluctuating hearing loss, tinnitus, aural fullness.	Salt restriction/diuretics and ENT-directed management.
Vestibular migraine	Episodic vertigo with migraine features; headache may be absent.	Migraine trigger control and prophylaxis when frequent.
Posterior circulation stroke	Vertigo with central signs, severe ataxia, vascular risk, abrupt persistent syndrome.	Stroke imaging and acute stroke pathway.

7. Neurocutaneous syndromes

Neurocutaneous syndromes connect visible skin findings with CNS, peripheral nerve, vascular, ocular, and tumor manifestations. They are high-yield because the skin clue may reveal the diagnosis before imaging. The uploaded source includes neurofibromatosis type 1, neurofibromatosis type 2, tuberous sclerosis, and Sturge-Weber syndrome.

Neurocutaneous syndromes

Syndrome	Skin / external clues	Neurologic and systemic associations
NF1	Cafe-au-lait macules, axillary/inguinal freckling, cutaneous neurofibromas, Lisch nodules.	Optic gliomas, CNS tumors, scoliosis, pheochromocytoma, learning disability, bony lesions.
NF2	May have fewer cafe-au-lait findings; bilateral vestibular schwannomas are classic.	Hearing loss, tinnitus, imbalance, meningiomas, ependymomas, cataracts.

Syndrome	Skin / external clues	Neurologic and systemic associations
Tuberous sclerosis	Hypomelanotic ash-leaf spots, facial angiofibromas, shagreen patches, periungual fibromas.	Seizures, developmental delay, cortical tubers, subependymal giant cell astrocytoma, renal angiomyolipomas, cardiac rhabdomyomas.
Sturge-Weber	Facial port-wine stain in trigeminal distribution.	Leptomeningeal angiomas, seizures, glaucoma, stroke-like episodes, developmental impairment.

8. Integrated cases

Case 1: painful monocular vision loss

A 28-year-old woman develops painful monocular vision loss, then months later has numbness below the chest after a hot shower.

Reasoning: Optic neuritis plus separated CNS symptoms suggests MS or another demyelinating disease. MRI brain/spine and selected CSF/serologic testing are appropriate.

Case 2: ascending weakness after diarrhea

A 36-year-old develops progressive leg weakness and areflexia 10 days after diarrheal illness. He has trouble counting to 20 in one breath.

Reasoning: GBS is likely. Admit, monitor respiratory mechanics and autonomic status, and treat significant disease with IVIG or plasma exchange.

Case 3: fluctuating diplopia and dysphagia

A 44-year-old has ptosis, diplopia, and chewing fatigue worse at the end of the day. Sensation and reflexes are normal.

Reasoning: Myasthenia gravis is likely. Test AChR/MuSK antibodies, consider EMG, image chest for thymoma, and monitor swallowing/respiration.

Case 4: cape-like sensory loss

A patient loses pain and temperature over shoulders and arms but has preserved vibration and position sense.

Reasoning: Syringomyelia affects central crossing spinothalamic fibers. MRI cervical spine and Chiari/tumor evaluation are needed.

Case 5: vertical nystagmus

A patient has continuous vertigo, vomiting, vertical nystagmus, dysarthria, and inability to walk.

Reasoning: Central vertigo from posterior circulation stroke or brainstem/cerebellar disease is likely. Do not treat as simple BPPV; pursue stroke evaluation.

Case 6: cafe-au-lait spots and optic glioma

A child has multiple cafe-au-lait macules, axillary freckling, learning difficulty, and an optic pathway tumor.

Reasoning: NF1 is likely; surveillance for vision, skeletal issues, blood pressure/pheochromocytoma concerns, and tumors is needed.

9. Graph-RAG extraction map

Concept nodes and relationships

Node	Class	Edges to encode
Multiple sclerosis	CNS demyelinating disease	presents_with optic neuritis/INO/relapsing deficits; diagnosed_by MRI/CSF oligoclonal bands; treated_by steroids for relapse and disease-modifying therapy.
Optic neuritis	Clinical syndrome	associated_with MS/NMOSD/MOGAD; presents_with painful monocular vision loss; evaluated_by MRI and ophthalmologic exam.
Guillain-Barre syndrome	Peripheral neuropathy	presents_with ascending weakness/areflexia; caused_after infection; complicated_by respiratory failure/autonomic instability; treated_by IVIG or plasma exchange.
Myasthenia gravis	Neuromuscular junction disorder	presents_with fatigable weakness/ptosis/diplopia/dysphagia; diagnosed_by antibodies/EMG; treated_by pyridostigmine/immunotherapy/thymectomy/IVIG/PLEX.

Node	Class	Edges to encode
Myasthenic crisis	Neurologic emergency	caused_by MG exacerbation; presents_with respiratory/bulbar weakness; treated_by ICU support and IVIG or plasma exchange.
Syringomyelia	Spinal cord disorder	causes cape-like pain-temperature loss; associated_with Chiari; diagnosed_by MRI.
Brown-Sequard syndrome	Spinal cord hemisection syndrome	causes ipsilateral weakness/proprioception loss and contralateral pain-temperature loss.
Transverse myelitis	Inflammatory spinal cord syndrome	presents_with bilateral deficits/sensory level/sphincter dysfunction; diagnosed_by MRI and CSF; treated_by steroids often.
Central vertigo	Brainstem/cerebellar syndrome	suggested_by vertical nystagmus/focal neurologic deficits/severe ataxia; evaluated_by stroke imaging.
NF1	Neurocutaneous syndrome	presents_with cafe-au-lait spots/neurofibromas/freckling; associated_with optic glioma/Lisch nodules/pheochromocytoma/skeletal findings.

10. High-yield traps

- Do not diagnose MS from nonspecific dizziness, fatigue, or tingling alone; require objective CNS lesions and exclusion of better explanations.
- A normal early CSF protein does not exclude Guillain-Barre syndrome.
- GBS can cause respiratory failure without early oxygen desaturation; monitor FVC and inspiratory force.
- Steroids are not useful for classic GBS, but are often used for MS relapse or transverse myelitis.
- Myasthenic crisis should be suspected before hypoxia; bulbar weakness and low FVC are major warnings.
- Avoid MG-worsening medications when possible, including aminoglycosides, fluoroquinolones, beta-blockers, magnesium, and some antiarrhythmics.
- A sensory level or bladder dysfunction points toward spinal cord disease, not simple peripheral neuropathy.
- Vertical or direction-changing nystagmus is central until proven otherwise.
- Neurocutaneous skin findings should prompt neurologic, ophthalmologic, renal, and tumor surveillance depending on the syndrome.

11. Practice questions

Q1. Painful monocular vision loss in a young adult with later relapsing sensory symptoms suggests what disease family?

Answer: CNS demyelinating disease such as multiple sclerosis, NMOSD, or MOG-associated disease.

Q2. Ascending weakness with areflexia after Campylobacter diarrhea suggests what?

Answer: Guillain-Barre syndrome.

Q3. What two treatments hasten recovery in significant GBS?

Answer: IVIG or plasma exchange.

Q4. Fatigable ptosis and diplopia with normal sensation suggest what?

Answer: Myasthenia gravis.

Q5. Which bedside respiratory concept matters in myasthenic crisis or GBS?

Answer: Ventilatory mechanics such as forced vital capacity and negative inspiratory force; oxygen saturation may remain normal until late.

Q6. Cape-like loss of pain and temperature over the shoulders suggests what?

Answer: Syringomyelia.

Q7. Ipsilateral weakness/proprioception loss with contralateral pain-temperature loss suggests what?

Answer: Brown-Sequard syndrome.

Q8. Vertigo with vertical nystagmus and dysarthria should be treated as what until proven otherwise?

Answer: Central vertigo/posterior circulation neurologic emergency.

Q9. Cafe-au-lait spots, axillary freckling, neurofibromas, and Lisch nodules suggest what?

Answer: Neurofibromatosis type 1.

12. References and clinical cross-check anchors

This chapter was generated as original educational content from the uploaded neurology source sections covering MS, GBS, myasthenia gravis, neurocutaneous syndromes, spinal cord diseases, and vertigo. Clinical anchors were cross-checked against AAN guidance for MS disease-modifying therapy, AAN thymectomy guidance for myasthenia gravis, international MG consensus guidance, and CDC/neurology references for Guillain-Barre syndrome. Local neurology protocols should guide disease-modifying therapy selection, respiratory thresholds, plasma exchange/IVIG access, and spinal cord emergency workflows.